When One Character Erases a Word: Orthographic Noise, Tokenizers and Negation in Arabic Sentiment Classification

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Abstract

Arabic tweets are full of orthographic variation that readers barely notice: the elongation character tatweel, emphatic letter repetition, and inconsistent spelling of hamza and final letters. We measure how this variation affects five Arabic and multilingual encoders fine-tuned for three-way sentiment classification on two tweet corpora, ArSarcasm-v2 and ArSAS, using about 40,000 controlled edits per model and three training seeds. At the input level the effect is stark: in all four Arabic WordPiece models, one tatweel turns the edited word into an unknown token, including every edited negation word. Predictions are sturdier than this suggests. A tatweel in a single word changes 3–7% of predictions, and the same edit in 30% of the words changes 9–18%. The extent to which a tokenizer destroys a word hardly predicts how often the prediction changes (Spearman $\rho \leq 0.18$). What differs is the cost: dense tatweel lowers Macro-F1 by 3.5–6.2 points in the Arabic models but by only 1.4–1.5 points in XLM-R. Writing tatweel on the negation word changes 2–3 points more predictions than writing it on another word in the same tweet. Two annotators judged every sampled tatweel edit to preserve meaning and sentiment, so these changes are errors. Removing tatweel and collapsing letter runs before tokenization undoes these edits at no measurable cost on clean text. A negation-aware normalizer we had proposed earlier adds nothing over the same rules without its negation guard. We release the perturbation suite, leakage-safe splits, predictions and the preregistered analysis plan.

1 Introduction

Social media Arabic is written with considerable freedom. Writers stretch a word with the tatweel character for visual effect (جميـــل for جميل "beautiful"), repeat a letter for emphasis (كثيييير, حلووو), and often omit hamza or write final ة as ه. In the two corpora we study, 3–5% of tweets contain a tatweel and a similar share repeat a letter three or more times (Table 7). Native readers glide over these variants. Whether sentiment classifiers do so is less clear, because the tokenizers that feed them were built from vocabularies in which these forms are rare or absent.

We arrived at this question through a negative result. A negation-aware normalization method we developed earlier did no better than a simpler normalizer under any condition. In that experiment, single-word edits applied to one MARBERT model produced accuracy changes too small to separate from zero. We concluded that the design, not the models, made the experiment uninformative. A single model gives no basis for comparison. A single edited word per tweet is a weak treatment. The repetition edits also skipped the long vowels ا, و and ي, which carry most emphatic repetition in real tweets.

This paper replaces that experiment with a broader, preregistered audit. We fine-tune five encoders with three seeds each on two tweet datasets. We then apply four families of orthographic edits at two densities, and in some conditions we edit the negation word itself. Two annotators judged whether the edits preserve meaning and sentiment. We ask four questions: (RQ1) how often do realistic edits change predictions; (RQ2) do tokenizer effects explain differences between models; (RQ3) is editing the negation word worse than editing another word; and (RQ4) does simple normalization before tokenization undo the damage?

The main findings are:

- Arabic WordPiece tokenizers map any word containing tatweel to `[UNK]`. This includes negation words, so an edited tweet such as لا reaches the model with its negation removed (§4).
- Predictions change far less than this input

damage would suggest: tatweel in one word changes 3–7% of them, and tatweel in 30% of the words changes 9–18%. The [UNK] rate barely predicts how many predictions change, but each WordPiece change costs about three times as much Macro-F1 as a change made by XLM-R (§6).

- Tatweel on the negation word changes more predictions than tatweel elsewhere in seven of eight model–dataset pairs. Elongating the negation word does not, but annotators read that elongation as an intensity change rather than noise (§7).

- Stripping tatweel and collapsing letter runs removes nearly all of these changes at no measurable cost on clean text. The negation guard of our earlier method adds nothing (§8).

2 Related work

Noise and robustness. Both synthetic and natural character noise degrade neural models (Belinkov and Bisk, 2018). Character-level recognizers have been proposed to defend against misspellings (Pruthi et al., 2019), and behavioural test suites probe whether a model is invariant to such edits (Ribeiro et al., 2020). Our edits are specific to Arabic script. We separate edits presumed to preserve meaning from a typo control and verify that separation with human judgments. We also report the costs of a defense alongside its benefits.

Arabic pretrained models. MARBERT (Abdul-Mageed et al., 2021), AraBERT (Antoun et al., 2020) and CAMeLBERT (Inoue et al., 2021) use WordPiece vocabularies learned from Arabic text. MARBERT and its successor share a vocabulary built from tweets. XLM-R (Conneau et al., 2020) uses a multilingual SentencePiece vocabulary that falls back on individual characters and therefore never produces an unknown token. The recommended preprocessing for some Arabic models removes tatweel, but nothing in the model interface enforces this. We are not aware of work that measures what omitting that step costs a fine-tuned model.

Negation. Benchmarks under-represent negation, and models trained on them often ignore it (Hossain et al., 2020). We do not parse negation scope. Instead, we compare an edit on the negation word with the same edit on another word in the same tweet.

Family	Original	Edited
Tatweel	حلو	حلـو
Elongation ×2	حلو	حلوو
Elongation ×4	كثير	كثيير
Folding	جميلة / أحب	جميله / احب
Typo (control)	جيد	جبد
Cue (tatweel)	لا	لـا

Table 1: The edit families, illustrated on single words.

3 Study design

We preregistered the conditions, metrics, hypotheses and decision rules before observing any model prediction on the new edits. The tokenizer audit in §4 was run before the preregistration and is disclosed there.

Data. ArSarcasm-v2 (Abu Farha et al., 2021) contains dialectal and MSA tweets labelled negative, neutral or positive. We use a leakage-safe split that removes from training any tweet whose normalized text occurs in the official test set (10,037 training and 2,509 validation tweets). ArSAS (Elmadany et al., 2018) contains 21,064 tweets with an additional Mixed class, which we drop. We merge repeated tweet IDs and identical texts into groups and exclude the 90 tweets whose groups carry conflicting labels, together with 44 further copies linked to excluded tweets. A stratified group split then gives 13,738 training, 1,966 validation and 3,923 test tweets. We considered ASTD (Nabil et al., 2015) as the second dataset and rejected it, because 20% of ArSarcasm-v2 training tweets (2,473) also occur in ASTD. ArSAS shares one tweet with ArSarcasm-v2, and that tweet is excluded. Each model is trained and evaluated separately on each dataset.

Edits. Table 1 illustrates the four families.

- *Tatweel* inserts one ـ after the last letter that joins to the next.

- *Elongation* repeats the last internal long vowel, or the final letter if there is none, two or four extra times.

- *Folding* writes أ, إ and آ as ا, final ة as ه, and final ى as ي.

- *Typo* replaces one interior letter with its neighbour on the Arabic keyboard.

Model	ArSarcasm-v2	ArSAS
MARBERT	74.7 ± 0.3	83.1 ± 0.3
MARBERTv2	75.8 ± 0.3	83.5 ± 0.2
AraBERT-tw	74.4 ± 0.2	83.3 ± 0.3
CAMeL-DA	71.0 ± 0.3	81.4 ± 0.3
XLM-R	71.0 ± 0.4	80.5 ± 0.2

Table 2: Clean validation Macro-F1 (%), mean $\pm$ standard deviation over three seeds.

The typo family was intended as a control that damages meaning. The human judgments in §5 show that it does so only partly.

Each family is applied at two densities: *single* edits one word, and *dense* edits $\lceil 0.3n \rceil$ of the n Arabic words in the tweet. In *cue* conditions the edit falls on an unambiguous negation word (لا, لم, لن, ليس, مش, مو, مافي, بدون and their variants). We leave out ما, whose relative, interrogative and exclamative uses outnumber its negating use. Non-cue conditions never edit a negation word. They also skip URLs, mentions, hashtags and tokens attached to Latin characters. Which word gets edited is decided by hashing the tweet ID and the condition. Single edits are a subset of the dense edits for the same tweet, and both elongation levels edit the same character. Per condition, edits apply to between 1,963 and 2,478 ArSarcasm-v2 tweets and between 1,657 and 1,951 ArSAS tweets. Cue conditions apply to the 404 and 316 tweets that contain an editable negation word.

Models. We fine-tune MARBERT, MARBERTv2, AraBERTv02-twitter, CAMeLBERT-DA and XLM-R-base with one fixed recipe:

- AdamW, learning rate 2×10^{-5}, batch size 16, at most five epochs;
- early stopping and checkpoint selection on validation Macro-F1;
- seeds 42, 43 and 44.

This gives 30 models. Revisions of all pretrained weights are pinned. Table 2 reports clean validation performance.

Defenses. At inference time we compare four preprocessors, each applied identically to clean and edited text:

- RAW leaves the input unchanged;
- STD removes tatweel and collapses runs of three or more identical letters to a single letter;
- NO_CTX and NASN2 are the two conservative normalizers from our earlier work.

NO_CTX and NASN2 share all of their rules except one: NASN2 also declines to collapse repeated letters near a negation word. Their repetition rule covers consonants only, so they do not undo elongated long vowels.

Metrics and statistics. The primary metric is the *flip rate*, the share of edited tweets whose predicted class differs from the prediction for the clean tweet. It needs no gold label and isolates the effect of the edit. We also report Macro-F1 on clean and edited versions of the same tweets and the difference between the two.

Intervals are 95% percentiles of 2,000 cluster-bootstrap resamples. Clusters are groups of duplicate texts, and the rate for each resample is averaged over the three seeds. Contrasts are paired on identical tweets. These include cue versus non-cue, dense versus single, a defense versus RAW, and each WordPiece model versus XLM-R. A hypothesis counts as supported for a model when its interval excludes zero in the stated direction.

Deviations from the preregistration. We report two deviations. First, the preregistered zero-shot LLM was not evaluated because the required GPU was unavailable. Second, this version reports the validation partitions only. The preregistered single evaluation on the held-out test partitions will be added once human adjudication is complete.

4 What the tokenizer sees

Figure 1 shows, for every model and edit family, how many tokens an edited word receives relative to the original. The pattern is identical on both datasets.

None of the four Arabic WordPiece vocabularies contains the tatweel character. WordPiece therefore replaces any word containing it with a single `[UNK]`. This happens in 100% of tatweel edits, including all 404 (ArSarcasm-v2) and 316 (ArSAS) edited negation words. A tweet containing ـ thus reaches these models with no negation. XLM-R never produces an unknown token. It instead splits a word containing tatweel into 2.5–2.6 times as many pieces, and an edited negation word into 2.9 times as many. Elongation fragments the word in every model: four extra letters multiply its token count by 2.5–4.1. Folding is largely absorbed by MARBERT, whose lowercasing strips hamza from

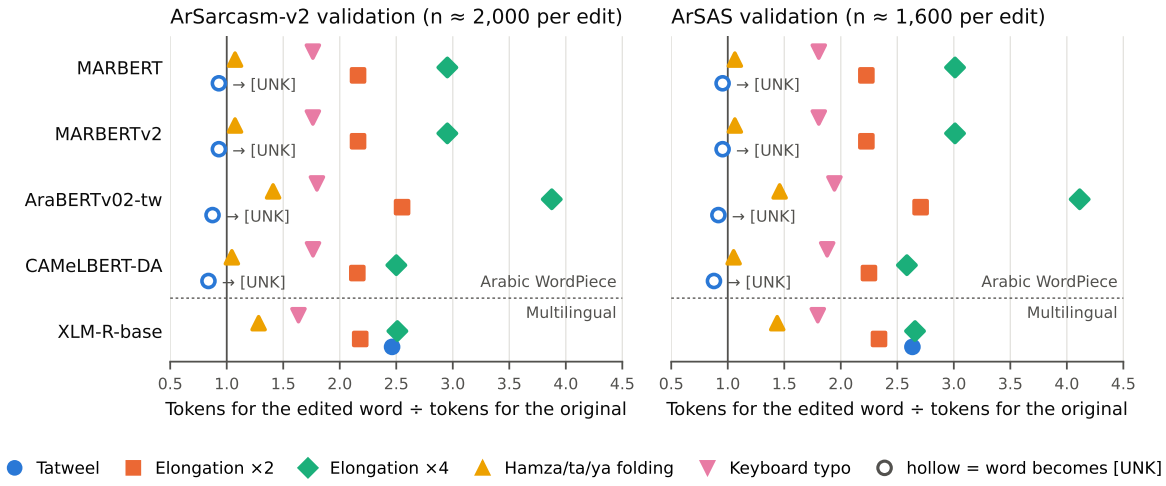


Figure 1: Number of tokens produced for an edited word, relative to the original word, averaged over single-word edits. A hollow marker means the edited word becomes `[UNK]` (rate 1.00 in every such cell). MARBERT and MARBERTv2 share a vocabulary, so their rows coincide.

Field	Agreement	κ	AC1
Meaning preserved	89.0%	0.43	0.88
Sentiment preserved	83.9%	0.56	0.80
Sentiment of edited text	80.3%	0.69	0.71
Negation present	92.9%	0.85	0.87
Negation affected by edit	91.3%	0.84	0.88

Table 3: Agreement between the two annotators on 310 items (20 pilot items excluded).

alef. As a result, 35.5% and 30.9% of its single-word folding edits leave the token sequence unchanged. The other models see nearly every folding edit.

5 Do humans notice the edits?

The study rests on the claim that most of these edits preserve meaning. To test it, two native-speaker annotators independently judged 330 edited tweets drawn from the ArSarcasm-v2 validation set, 30 per condition and one edit per tweet. They answered five questions per item. Their first 20 items served as a pilot and are excluded from the agreement figures. Agreement (Table 3) is high on every field. Cohen's κ is low for meaning preservation because almost every answer is "yes". Gwet's AC1 (Gwet, 2008) corrects for this skew and gives 0.88. A third annotator resolved the 87 items on which the two disagreed about meaning or sentiment preservation. That annotator saw both earlier answers, without the annotators' identities or notes and in random order. The third annotator sided with each of the two about equally often on meaning (76% and 75%). On sentiment, the third annotator sided more often with annotator A (52%) than with annotator B (21%), who had marked many of these items unsure.

The results (Table 4) separate the families clearly.

- *Tatweel* preserved both meaning and sentiment in all 90 items, including the 30 in which it fell on a negation word. Any prediction change under tatweel is therefore an error.

- *Elongation* preserved meaning in all 120 items, but it often changed sentiment. Both annotators judged the sentiment unchanged in only 17 and 14 of 30 items for one-word elongation by two and four letters, in 6 of 30 when 30% of the words were elongated, and in none of the 30 items with an elongated negation word. Readers take repetition as emphasis, so a prediction change under elongation is not necessarily an error.

- *Folding* preserved meaning in 48 of 60 items by agreement, and in all 60 after adjudication. The disputed items wrote على as علي or dropped a hamza, which the third annotator judged cosmetic.

- *Typos*, meant as the control, were often read through. After adjudication, 19 of 30 single-word typos and 8 of 30 dense ones preserved meaning and sentiment.

Condition	Meaning (A / B)	Both: meaning	Both: meaning + sentiment	Final
Tatweel, 1 word	30 / 30	30	30	30
Tatweel, 30% of words	30 / 30	30	30	30
Tatweel on negation cue	30 / 30	30	30	30
Elongation $\times 2$, 1 word	30 / 30	30	17	17
Elongation $\times 4$, 1 word	30 / 30	30	14	14
Elongation $\times 2$, 30% of words	30 / 30	30	6	6
Elongation $\times 2$ on negation cue	30 / 30	30	0	0
Folding, 1 word	30 / 25	25	25	30
Folding, 30% of words	29 / 24	23	23	30
Typo (control), 1 word	24 / 20	18	18	19
Typo (control), 30% of words	13 / 11	8	8	8

Table 4: Items, out of 30 per condition, judged meaning-preserving by annotator A and by annotator B; items for which both judged meaning preserved, or both meaning and sentiment preserved; and the final count of meaning- and sentiment-preserving items after a third annotator resolved the disagreements.

Adjudication changed none of the counts for tatweel or elongation. Every elongation item it resolved was judged meaning-preserving but sentiment-changing.

6 How often do predictions change?

Figure 2 shows the flip rates. A single edited word rarely matters.

- Tatweel in one word changes 5.0–6.6% of predictions on ArSarcasm-v2 and 3.0–3.9% on ArSAS.
- Two extra letters change 2.5–6.8%.
- Folding changes 0.9–3.6%.

Density matters a great deal. When 30% of the words carry tatweel, flip rates rise to 11.8–17.5% on ArSarcasm-v2 and 8.6–10.9% on ArSAS. Dense elongation reaches 18.6% for XLM-R. Every dense-versus-single contrast for the three presumed-preserving families is positive, in all five models and on both datasets (30 of 30; H1). The increases range from 1.1 to 3.0 points for folding and from 5.5 to 11.9 points for tatweel and elongation.

Tokenizer damage barely predicts flip rates. The tokenizer audit predicts a large gap between WordPiece models, which lose the edited word entirely, and XLM-R, which keeps a fragmented version. The gap is real but small. Under dense tatweel, every WordPiece model flips more often than XLM-R on ArSarcasm-v2, by 3.1–5.7 points. On ArSAS only two of the four do so, by 1.6 and 1.7 points (H2: 6 of 8 contrasts). On ArSarcasm-v2, XLM-R is in fact the most sensitive model to dense elongation and to dense typos.

Across the 55 model–condition cells per dataset, the share of edited words that become `[UNK]` correlates only weakly with flip rate (Spearman $\rho = 0.18$ and 0.14), and fragmentation does not correlate at all ($\rho \leq 0.03$). One explanation is that fine-tuning already exposes the models to `[UNK]` for tatweel words: 4.6% of ArSarcasm-v2 training and validation tweets contain a natural tatweel.

But WordPiece errors cost more. The flip rate does not say whether a changed prediction is wrong. Table 5 does. Under dense tatweel, Macro-F1 falls by 4.3–6.2 points for the WordPiece models on ArSarcasm-v2 and by 3.5–4.0 points on ArSAS. For XLM-R it falls by only 1.5 and 1.4 points. Per changed prediction, a WordPiece model loses about three times as much Macro-F1 as XLM-R. On ArSarcasm-v2, for example, MARBERT loses 0.38 Macro-F1 points per point of flip rate, against 0.13 for XLM-R. This is consistent with an erased sentiment word removing the evidence for the label rather than merely adding noise. Dense typos cost the most: 5.4–10.9 points across all models.

7 Negation as a stress test

Natural noise on a negation word is rare. In the training and validation tweets, at most five per corpus contain an elongated or tatweel-marked negation word. We therefore treat the cue conditions as a controlled probe of how models represent negation, not as an estimate of real-world risk.

For each tweet with an editable negation word, we compare editing that word with editing another word in the same tweet (H3). Tatweel on the negation word, which turns it into `[UNK]`, changes more predictions in seven of the eight WordPiece model–dataset pairs. The increase is 2.4–3.3 points on

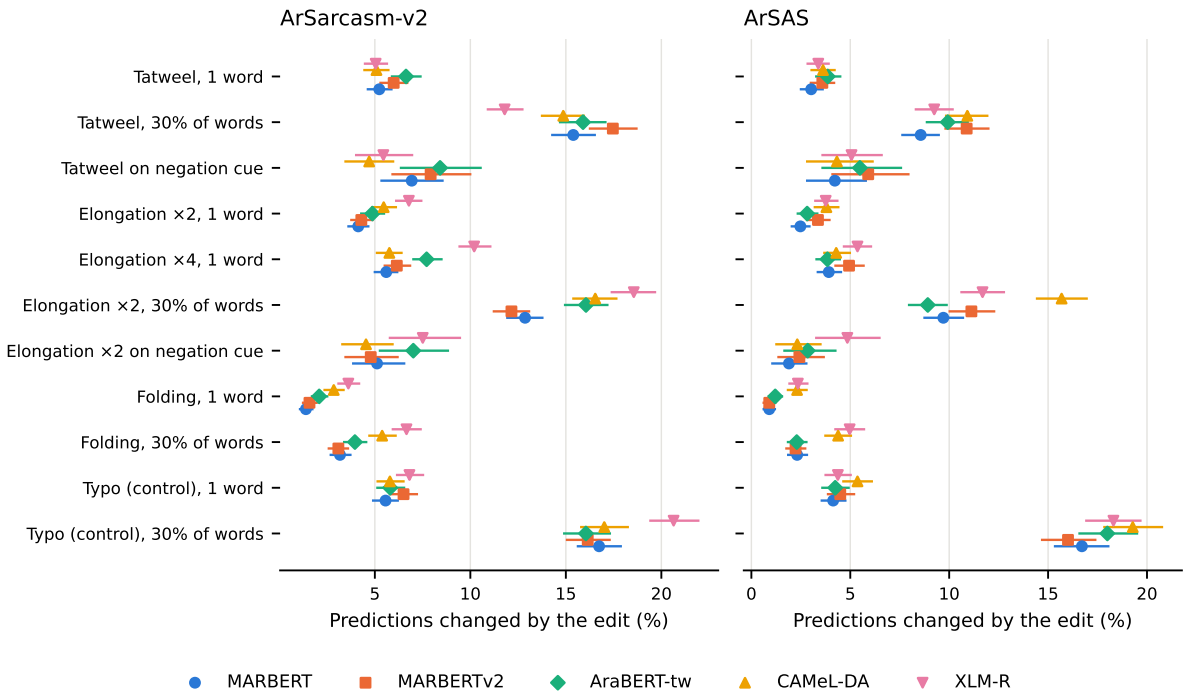


Figure 2: Flip rates with 95% cluster-bootstrap intervals (RAW inputs, three seeds averaged per model). The full tables are in Appendix A.

	ArSarcasm-v2			ArSAS		
	Tatweel	Elong. ×2	Typo	Tatweel	Elong. ×2	Typo
MARBERT	5.9 [4.6, 7.2]	4.4 [3.4, 5.3]	7.5 [6.2, 8.7]	3.5 [2.5, 4.6]	4.6 [3.6, 5.6]	10.9 [9.4, 12.3]
MARBERTv2	6.2 [4.8, 7.7]	2.9 [1.9, 3.9]	5.4 [4.1, 6.7]	4.0 [2.8, 5.2]	4.3 [3.2, 5.5]	7.3 [5.8, 8.8]
AraBERT-tw	5.0 [3.6, 6.3]	3.7 [2.4, 4.9]	5.4 [3.9, 6.8]	3.7 [2.6, 4.9]	3.3 [2.3, 4.3]	9.8 [8.2, 11.4]
CAMeL-DA	4.3 [2.9, 5.6]	4.2 [3.0, 5.5]	5.4 [4.0, 6.9]	3.7 [2.6, 4.8]	6.1 [4.7, 7.5]	9.2 [7.6, 10.7]
XLM-R	1.5 [0.6, 2.5]	5.5 [4.4, 6.7]	7.5 [6.1, 8.9]	1.4 [0.5, 2.4]	3.6 [2.6, 4.8]	7.1 [5.6, 8.6]

Table 5: Macro-F1 drop in points (clean minus edited, same tweets) under dense edits, with 95% intervals.

ArSarcasm-v2 and 2.0–3.1 points on ArSAS. The exception is CAMeLBERT-DA on ArSarcasm-v2 (+0.3, interval spanning zero). The effect is consistent but modest. Losing the negation word changes the predicted class of roughly one additional tweet in thirty to forty, so these models evidently do not rest their decisions on the negation word alone.

Elongating the negation word (ﻻ) increases flip rates significantly in only two of ten model–dataset pairs. The annotators read this edit very differently from tatweel. Annotator A judged the sentiment changed in 26 of 30 items, and annotator B in 16 (with 13 marked unsure). In no item did both annotators judge it unchanged (§5). An elongated ﻻ is an emphatic refusal, and a model that changes its prediction may be reading it correctly.

8 Does normalization help?

Table 6 compares the defenses. STD removes tatweel and collapses letter runs, and so reverses exactly the edits it targets. It brings flip rates under tatweel to zero and under dense elongation to at most 0.2% (H4: 70 of 70 tests). It does not touch folding or typos.

The more relevant question is what STD costs on unedited text, where natural letter runs may carry emphasis. The cost is negligible. On the clean validation tweets used in the tatweel conditions (2,478 and 1,951), Macro-F1 under STD differs from RAW by at most 0.15 points for every model and dataset.

The two conservative normalizers from our earlier work remove most tatweel effects. They leave 0.5–1.0% of flips, because they keep tatweel inside the words they treat as negation markers, which is a broader set than our cue list. They do not undo elongated long vowels, which their repeti-

Edit	Model	ArSarcasm-v2 RAW	STD	NO_CTX	NASN2	ArSAS RAW	STD	NO_CTX	NASN2
Tatweel	MARBERT	15.4	0.0	0.8	0.8	8.6	0.0	0.6	0.6
	MARBERTv2	17.5	0.0	0.9	0.9	10.9	0.0	0.9	0.9
	AraBERT-tw	15.9	0.0	1.0	1.0	9.9	0.0	0.9	0.9
	CAMeL-DA	14.9	0.0	0.7	0.7	10.9	0.0	0.5	0.5
	XLM-R	11.8	0.0	0.8	0.8	9.2	0.0	0.7	0.7
Elongation $\times 2$	MARBERT	12.9	0.1	12.2	12.2	9.7	0.0	9.7	9.6
	MARBERTv2	12.2	0.2	11.6	11.6	11.1	0.1	10.3	10.3
	AraBERT-tw	16.1	0.2	15.4	15.5	8.9	0.2	8.7	8.7
	CAMeL-DA	16.5	0.2	15.8	15.9	15.7	0.1	15.1	15.0
	XLM-R	18.6	0.2	17.4	17.5	11.7	0.2	11.6	11.5
Folding	MARBERT	3.2	3.1	3.1	3.1	2.3	2.3	2.4	2.4
	MARBERTv2	3.1	3.1	3.1	3.1	2.2	2.3	2.2	2.2
	AraBERT-tw	4.0	4.0	3.9	3.9	2.3	2.3	2.3	2.3
	CAMeL-DA	5.4	5.3	5.4	5.4	4.4	4.3	4.3	4.3
	XLM-R	6.7	6.5	6.6	6.6	5.0	5.1	5.0	5.0

Table 6: Flip rates (%) under dense edits for each preprocessing defense, applied to both clean and edited text.

	ArSarcasm-v2	ArSAS
Tweets (train + val.)	12,546	15,704
with tatweel	571 (4.6%)	502 (3.2%)
with a letter run ≥ 3	498 (4.0%)	436 (2.8%)
of which a long vowel	300 (2.4%)	294 (1.9%)
tatweel on a negation word	1	1
letter run on a negation word	5	3

Table 7: Naturally occurring noise. Test partitions were not read.

tion rule excludes. The negation guard that distinguishes NASN2 from NO_CTX changes nothing that matters. Across 110 paired contrasts the two differ by at most 0.2 points. None favours NASN2, and five favour NO_CTX by fractions of a point. Under the preregistered rule, the claim that negation-aware normalization improves robustness is not supported (H5).

9 Natural noise

Controlled edits show what can happen; natural noise shows what does (Table 7). For the validation tweets that already contain noise, we compare the prediction for the tweet as written with the prediction after STD cleaning (Table 8). Cleaning a natural tatweel changes 0–5% of predictions. Collapsing natural letter runs changes 11–15% on ArSarcasm-v2 and 2–8% on ArSAS. The Macro-F1 differences on these slices are not significant: they contain only 42–123 tweets, and natural repetition carries genuine emphasis that cleaning removes. Natural noise thus moves a noticeable minority of predictions, but this sample is too small to show whether cleaning makes them more accurate.

10 Discussion

The tokenizer view and the prediction view tell different stories. A WordPiece model receives an `[UNK]` in place of every tatweel word, which sounds catastrophic. Yet the prediction usually survives, because sentiment in a tweet rarely depends on a single word. Where the two views agree is in the cost of the predictions that do change. When an Arabic WordPiece model loses a word and changes its prediction, the change costs about three times as much Macro-F1 as a change made by XLM-R, which merely fragments the word.

The families also call for different readings. Tatweel is pure typography. Annotators saw no change in 90 of 90 items, so every prediction change under tatweel is an error, and a trivial fix removes it. Elongation is not noise in the same sense: readers took it as a change of intensity in most items, and a model that responds to it is not necessarily wrong. Benchmarks that treat all orthographic variation as a meaning-preserving perturbation conflate these cases.

The practical recommendation follows directly. Anyone deploying an Arabic WordPiece model on social media text should remove tatweel before tokenization. The cost on clean text was below 0.15 Macro-F1 points in every model we tested. Collapsing letter runs is equally cheap, but should be a deliberate choice, since it also removes emphasis. Negation-aware exceptions to these rules did not help in our experiments.

Model	ArSarcasm-v2		ArSAS	
	Tatweel (n=123)	Letter run $\geq$3 (n=97)	Tatweel (n=51)	Letter run $\geq$3 (n=42)
MARBERT	2.2 [0.5, 4.1]	14.8 [9.3, 20.3]	5.2 [1.3, 9.8]	7.9 [3.2, 13.6]
MARBERTv2	2.7 [0.8, 5.4]	11.3 [6.2, 16.8]	2.0 [0.0, 4.6]	7.9 [2.3, 14.6]
AraBERT-tw	3.0 [1.1, 5.1]	13.4 [8.2, 19.2]	0.0 [0.0, 0.0]	2.4 [0.0, 6.5]
CAMeL-DA	2.7 [0.8, 5.1]	13.1 [7.6, 18.9]	3.3 [0.7, 7.2]	2.4 [0.0, 6.5]
XLM-R	2.4 [0.5, 4.6]	14.1 [9.3, 19.6]	2.0 [0.0, 4.6]	5.6 [0.8, 11.9]

Table 8: Validation tweets that already contain noise: flip rate (%) between the tweet as written and its STD-cleaned form, with 95% intervals.

Limitations

The edits are rule-based and applied one family at a time, whereas real noise co-occurs. The negation lexicon is finite and leaves out لم. Checkpoints were selected on the validation partitions, which favours clean accuracy but not the flip rate. The human validation covers one dataset and 30 items per condition. Both datasets consist of tweets, and other genres may behave differently. The preregistered LLM comparison and the test-partition evaluation are not yet reported.

Ethics statement

We use public research datasets under their licenses and release only tweet IDs, hashes, labels and aggregate results, never tweet text. Two expert annotators labelled the human-validation sample.

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A Full results tables

Tables 9 and 10 list the flip rate for every condition and model. Table 11 counts the supported tests for each preregistered hypothesis. Per-test estimates and intervals are in the released bundle.

B Reproducibility

Code, configurations and text-free artefacts are released with the paper. The artefacts include perturbation logs, split manifests, predictions keyed by text hash, bootstrap results and the preregistration. Every table in this paper is generated from the released analysis bundle by a script; no number was entered by hand. Training is deterministic for a fixed seed, GPU type and software stack, and checkpoint hashes are published so that regenerated weights can be verified.

Condition	MARBERT	MARBERTv2	AraBERT-tw	CAMeL-DA	XLM-R
Tatweel, 1 word	5.2	6.0	6.6	5.1	5.0
Tatweel, 30% of words	15.4	17.5	15.9	14.9	11.8
Tatweel on negation cue	6.9	7.9	8.4	4.7	5.4
Elongation $\times 2$, 1 word	4.1	4.3	4.9	5.5	6.8
Elongation $\times 4$, 1 word	5.6	6.1	7.7	5.8	10.2
Elongation $\times 2$, 30% of words	12.9	12.2	16.1	16.5	18.6
Elongation $\times 2$ on negation cue	5.1	4.8	7.0	4.5	7.5
Folding, 1 word	1.4	1.6	2.1	2.8	3.6
Folding, 30% of words	3.2	3.1	4.0	5.4	6.7
Typo (control), 1 word	5.6	6.5	5.8	5.8	6.8
Typo (control), 30% of words	16.7	16.1	16.1	17.0	20.6

Table 9: ArSarcasm-v2 validation: flip rate (%) for every condition, RAW inputs.

Condition	MARBERT	MARBERTv2	AraBERT-tw	CAMeL-DA	XLM-R
Tatweel, 1 word	3.0	3.6	3.9	3.6	3.4
Tatweel, 30% of words	8.6	10.9	9.9	10.9	9.2
Tatweel on negation cue	4.2	5.9	5.5	4.3	5.1
Elongation $\times 2$, 1 word	2.5	3.4	2.8	3.8	3.8
Elongation $\times 4$, 1 word	3.9	4.9	3.8	4.3	5.4
Elongation $\times 2$, 30% of words	9.7	11.1	8.9	15.7	11.7
Elongation $\times 2$ on negation cue	1.9	2.4	2.8	2.3	4.9
Folding, 1 word	0.9	0.9	1.2	2.3	2.4
Folding, 30% of words	2.3	2.2	2.3	4.4	5.0
Typo (control), 1 word	4.1	4.5	4.2	5.4	4.4
Typo (control), 30% of words	16.7	16.0	18.0	19.3	18.3

Table 10: ArSAS validation: flip rate (%) for every condition, RAW inputs.

Hypothesis	ArSarcasm-v2	ArSAS
H1	15/15	15/15
H2	4/4	2/4
H3	3/4	4/4
H4	35/35	35/35
H5	0/55	0/55

Table 11: Supported tests per preregistered hypothesis (95% interval excludes zero in the stated direction).